

# Analysis

PENG

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## 1 Introduction

Mathematics treats its objects abstractly, caring only about what **properties** the objects have [1].

## 2 Natural Number system

### 2.1 The Peano Axioms

**Axiom 2.1.** *0 is a natural number.*

**Axiom 2.2.** *If  $n$  is a natural number, then  $n++$  is also a natural number.*

**Definition 2.3.**  $1 := 0++$ ,  $2 := 1++$ ,  $3 := 2++$ ,  $\dots$

**Proposition 2.4.** *3 is a natural number.*

*Proof.* By Axiom 2.1, 0 is a natural number.

By Axiom 2.2,  $0++ = 1$  is a natural number.

By Axiom 2.2,  $1++ = 2$  is a natural number.

By Axiom 2.2,  $2++ = 3$  is a natural number. □

**Axiom 2.5.**  $n++ \neq 0$

**Proposition 2.6.**  $4 \neq 0$

*Proof.* By definition 2.3,  $3++ = 4$ .

By Proposition 2.4, 3 is a natural number.

By Axiom 2.5,  $3++ = 4 \neq 0$ . □

**Axiom 2.7.**  $n \neq m \Rightarrow n++ \neq m++$  or  $n++ = m++ \Rightarrow n = m$

**Axiom 2.8** (principle of mathematical induction). *Suppose that  $P(0)$  is true, and suppose that whenever  $P(n)$  is true,  $P(n++)$  is also true. Then  $P(n)$  is true for every natural number  $n$ .*

## 2.2 Addition

**Definition 2.9** (Addition). *Let  $n$  and  $m$  be natural numbers.*

1.  $0 + m := m$
2.  $(n++) + m := (n + m)++$

**Proposition 2.10.** *Let  $n$  and  $m$  be natural numbers. Then  $n + m$  is also a natural number.*

**Lemma 2.11.** *Let  $n$  and  $m$  be natural numbers.*

1.  $n + 0 = n$
2.  $n + (m++) = (n + m)++$

**Proposition 2.12.** *Let  $a, b, c$  be natural numbers.*

1.  $a + b = b + a$
2.  $(a + b) + c = a + (b + c)$
3.  $a + b = a + c \Rightarrow b = c$

**Definition 2.13.** *A natural number  $n$  is positive iff (if and only if)  $n \neq 0$ .*

**Proposition 2.14.** *Let  $a$  and  $b$  be natural numbers.*

1. *If  $a$  is positive, then  $a + b$  is positive.*
2.  $a + b = 0 \Rightarrow a = b = 0$ .
3. *If  $a$  is positive, then there exists only one natural number  $b$  such that  $b++ = a$ .*

*Proof.* Prop. 2.14 (1).

$a + 0 = a$  is positive. Suppose that  $a + b$  is positive. Then  $a + b++ = (a + b)++ \neq 0$  is positive.  $\square$

*Proof.* Prop. 2.14 (3).

Suppose that  $a = b++$ . Then  $a++ = (a)++$ .  $\square$

**Definition 2.15.** *Let  $n$  and  $m$  be natural numbers.*

1.  $n \geq m$  or  $m \leq n$  iff  $n = m + a$
2.  $n > m$  or  $m < n$  iff  $n \geq m$  and  $n \neq m$

**Proposition 2.16.** *Let  $a, b, c$  be natural numbers.*

1.  $a \geq a$
2. *If  $a \geq b$  and  $b \geq c$ , then  $a \geq c$*
3. *If  $a \geq b$  and  $b \geq a$ , then  $a = b$*
4.  $a \geq b$  iff  $a + c \geq b + c$
5.  $a < b$  iff  $a++ \leq b$
6.  $a < b$  iff  $b = a + d$  for some positive number  $d$

*Proof.* Prop. 2.16 (5).

$b = a + d$  and  $b \neq a \Rightarrow d \neq 0 \Rightarrow d = n++$

$b = a + (n++) = (a++) + n \Rightarrow a++ \leq b$   $\square$

**Proposition 2.17.** *Let  $a$  and  $b$  be natural numbers. Then exactly one of following statement is true:  $a < b$ ,  $a = b$ , or  $a > b$ .*

*Proof.* If  $a < b$  and  $a > b \xrightarrow{\text{Prop. 2.16 (3)}} a = b$  contradiction  $\square$

**Proposition 2.18** (Strong principle of induction). *Let  $m_0$  be a natural number. Suppose that for each  $m \geq m_0$ , if  $P(m')$  is true for all  $m_0 \leq m' < m$ , then  $P(m)$  is also true. Then  $P(m)$  is true for all  $m \geq m_0$ .*

*Proof.* Let  $Q(n)$  be the property that  $P(m)$  is true for all  $m_0 \leq m < n$ .

Suppose that if  $Q(n)$  is true, then  $P(n)$  is also true.

Then  $P(m)$  is true for all  $m_0 \leq m < n++$ . □

**Example 2.19.** *Fibonacci numbers.*

$$F_n = \frac{\phi^n - \psi^n}{\phi - \psi} \quad (2.1)$$

$$F_{n+2} = F_{n+1} + F_n \quad (2.2)$$

where  $\phi = \frac{1+\sqrt{5}}{2}$  and  $\psi = \frac{1-\sqrt{5}}{2}$

**Exercise 2.20.** *Principle of backwards induction.*

*Let  $n$  be a natural number. Whenever  $P(m++)$  is true,  $P(m)$  is true. If  $P(n)$  is true, then  $P(m)$  is true for all  $m \leq n$ .*

*Proof.* Suppose that if  $P(n)$  is true, then  $P(m)$  is true for all  $m \leq n$ . If  $P(n++)$  is true, then  $P(m)$  is true for all  $m \leq n++$ . □

**Exercise 2.21.** *Principle of induction starting from the base case  $n$ .*

*Let  $n$  be a natural number. Whenever  $P(m)$  is true,  $P(m++)$  is true. If  $P(n)$  is true, then  $P(m)$  is true for all  $m \geq n$ .*

*Proof.* If  $P(0)$  is true, suppose that  $P(m)$  is true. Then  $P(m++)$  is also true. Then  $P(m)$  is true for all  $m \geq 0$ . Suppose that if  $P(n)$  is true, then  $P(m)$  is true for all  $m \geq n$ . If  $P(n++)$  is true, then  $P(m)$  is true for all  $m \geq n++$ . □

## 2.3 Multiplication

**Definition 2.22.** *Let  $n$  and  $m$  be natural numbers.*

1.  $0 \times m := 0$
2.  $(n++) \times m := (n \times m) + m$

**Proposition 2.23.** *Let  $n$  and  $m$  be natural numbers. Then  $n \times m$  is also a natural number.*

**Lemma 2.24.** *Let  $n$  and  $m$  be natural numbers.*

1.  $n \times 0 = 0 \times n$
2.  $n \times (m++) = (n \times m) + n$

**Lemma 2.25** (Commutative). *Let  $n$  and  $m$  be natural numbers.*

1.  $n \times m = m \times n$
2.  $n \times m = 0$  iff at least one of  $n$  and  $m$  is equal to 0. If  $n$  and  $m$  are both positive, then  $nm$  is also positive.

**Proposition 2.26.** *Let  $a, b, c$  be natural numbers.*

1.  $a(b+c) = (b+c)a = bc+ca$
2.  $(ab)c = a(bc)$
3. If  $a < b$  and  $c$  is positive, then  $ac < bc$ .
4. If  $ac = bc$  and  $c$  is positive, then  $a = b$ .

**Lemma 2.27** (Euclid's division lemma). *Let  $n$  be a natural number and  $q$  be a positive natural number. Then there exist natural numbers  $m$  and  $r$ , such that  $0 \leq r < q$  and  $n = mq + r$ .*

*Proof.*  $0 = 0 \times q + 0$  and  $0 \leq r < q$ . Suppose that  $n = mq + r$  and  $0 \leq r < q$ . Then  $n++ = mq + (r++)$  and  $0 \leq r++ \leq q$ . If  $r++ = q$ ,  $n++ = mq + q = (m+1)q + 0$ . If  $r++ < q$ ,  $n++ = mq + r++$ .  $\square$

**Definition 2.28** (Exponentiation).

1.  $m^0 := 1$  and  $0^0 := 1$
2.  $m^{n++} := m^n \times m$

### 3 Set Theory

**Definition 3.1.** Set  $A$  is any unordered collection of objects.  $x \in A$  if  $x$  lies in the collection, otherwise  $x \notin A$ .

**Axiom 3.2.**

1. If  $A$  is a set, then  $A$  is also an object.
2.  $A = B$  iff every element of  $A$  belongs also to  $B$  and vice versa.
3. There exists an empty set  $\emptyset$ , such that every object  $x \notin \emptyset$ .
4. Let  $A$  be a non-empty set. Then there exist an object  $x \in A$ .
5. Singleton set  $\{a\}$ :  $y \in \{a\}$  iff  $y = a$ . Pair set  $\{a, b\}$ :  $y \in \{a, b\}$  iff  $y = a$  or  $y = b$ .
6.  $x \in A \cup B \Leftrightarrow x \in A$  or  $x \in B$ .
7.  $y \in \{x \in A : P(x) \text{ is true}\} \Leftrightarrow y \in A$  and  $P(y)$  is true.
8.  $z \in \{y : P(x, y) \text{ is true for some } x \in A\} \Leftrightarrow P(x, z) \text{ is true for some } x \in A$ .
9. There exists a set  $\mathbb{N}$ , whose elements are called natural numbers, and an object  $n++$  assigned to every  $n \in \mathbb{N}$ , such that the Peano axioms ?? hold.

**Example 3.3.**  $\{a, a\} = \{a\}$

**Exercise 3.4.**  $\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}$  are all distinct.

*Proof.*  $y \notin \emptyset$ .

$$y = \emptyset \in \{\emptyset\}.$$

$$y = \{\emptyset\} \in \{\{\emptyset\}\}.$$

$$y = \emptyset \text{ or } y = \{\emptyset\} \in \{\emptyset, \{\emptyset\}\}.$$

$\square$

**Lemma 3.5.** Let  $a, b$  are objects, and  $A, B, C$  are sets.

1.  $\{a, b\} = \{a\} \cup \{b\}$
2.  $A \cup B = B \cup A$
3.  $(A \cup B) \cup C = A \cup (B \cup C)$

**Definition 3.6.**

$$\begin{aligned} \{a, b, c\} &:= \{a\} \cup \{b\} \cup \{c\} \\ \{a, b, c, d\} &:= \{a\} \cup \{b\} \cup \{c\} \cup \{d\} \\ &\vdots \end{aligned}$$

**Definition 3.7** (Subsets). 1.  $A \subseteq B \Leftrightarrow$  for any object  $x \in A$ , then  $x \in B$ .

2.  $A \subsetneq B$  if  $A \subseteq B$  and  $A \neq B$ .

**Proposition 3.8.** Let  $A, B, C$  be sets.

1. If  $A \subseteq B$  and  $B \subseteq C$ , then  $A \subseteq C$ .

2. If  $A \subseteq B$  and  $B \subseteq A$ , then  $A = B$ .

3. If  $A \subsetneq B$  and  $B \subsetneq C$ , then  $A \subsetneq C$ .

**Definition 3.9.** Let  $A, B$  be sets.

1.  $A \cap B := \{x \in A : x \in B\}$ .

2.  $A \setminus B := \{x \in A : x \notin B\}$ .

**Proposition 3.10.** Let  $A, B, C$  be sets, and  $\{A, B, C\} \subseteq X$ .

1.  $A \cup \emptyset = A$  and  $A \cap \emptyset = \emptyset$ .

2.  $A \cup X = X$  and  $A \cap X = A$ .

3.  $A \cup A = A$  and  $A \cap A = A$ .

4.  $A \cup B = B \cup A$  and  $A \cap B = B \cap A$ .

5.  $(A \cup B) \cup C = A \cup (B \cup C)$  and  $(A \cap B) \cap C = A \cap (B \cap C)$ .

6.  $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$  and  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ .

7.  $A \cup (X \setminus A) = X$  and  $A \cap (X \setminus A) = \emptyset$ .

8. De Morgan laws:  $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$  and  $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$ .

*Proof.* Prop. 3.10 (5).

$$x \in A \cap X \Leftrightarrow x \in A \Rightarrow A \cap X = A. \quad \square$$

*Proof.* Prop. 3.10 (6).

$$x \in A \cap (B \cup C) \Rightarrow x \in A \text{ and } (x \in B \text{ or } x \in C) \Rightarrow x \in A \cap B \text{ or } x \in A \cap C \Rightarrow x \in (A \cap B) \cup (A \cap C).$$

$$x \in (A \cap B) \cup (A \cap C) \Rightarrow x \in (A \cap B) \text{ or } x \in (A \cap C) \Rightarrow x \in A \text{ and } x \in B \cup C \Rightarrow x \in A \cap (B \cup C). \quad \square$$

**Lemma 3.11** (De Morgan laws).

$$\neg(A \text{ or } B) = \neg A \text{ and } \neg B.$$

$$\neg(A \text{ and } B) = \neg A \text{ or } \neg B.$$

*Proof.* Prop. 3.10 (8).

$$x \in X \setminus (A \cup B) \Rightarrow x \in X \text{ and } (x \notin A \text{ and } x \notin B).$$

$$x \in X \setminus (A \cap B) \Rightarrow x \in X \text{ and } (x \notin A \text{ or } x \notin B). \quad \square$$

**Example 3.12.**

$$\{y : y = f(x) \text{ for some } x \in A\} = \{f(x) : x \in A\}.$$

$$\{x \in A : P(x) \text{ is true}\} \Rightarrow \{f(x) : x \in A; P(x) \text{ is true}\}.$$

**Example 3.13.**  $\{n + 3 : n \in \mathbb{N}, 0 \leq n \leq 5\} = \{8 - m : m \in \mathbb{N}, 0 \leq m \leq 3\}$ .

**Exercise 3.14** (Axiom 3.2 (8)  $\Rightarrow$  Axiom 3.2 (7)).

$$y \in \{x : P(x) \text{ is true for some } x \in A\} \Rightarrow y \in \{x \in A : P(x) \text{ is true}\}.$$

## 3.1 Functions

## References

[1] Terence Tao. *Analysis I*. Vol. 1. Springer, 2022. ISBN: 978-981-19-7261-4.